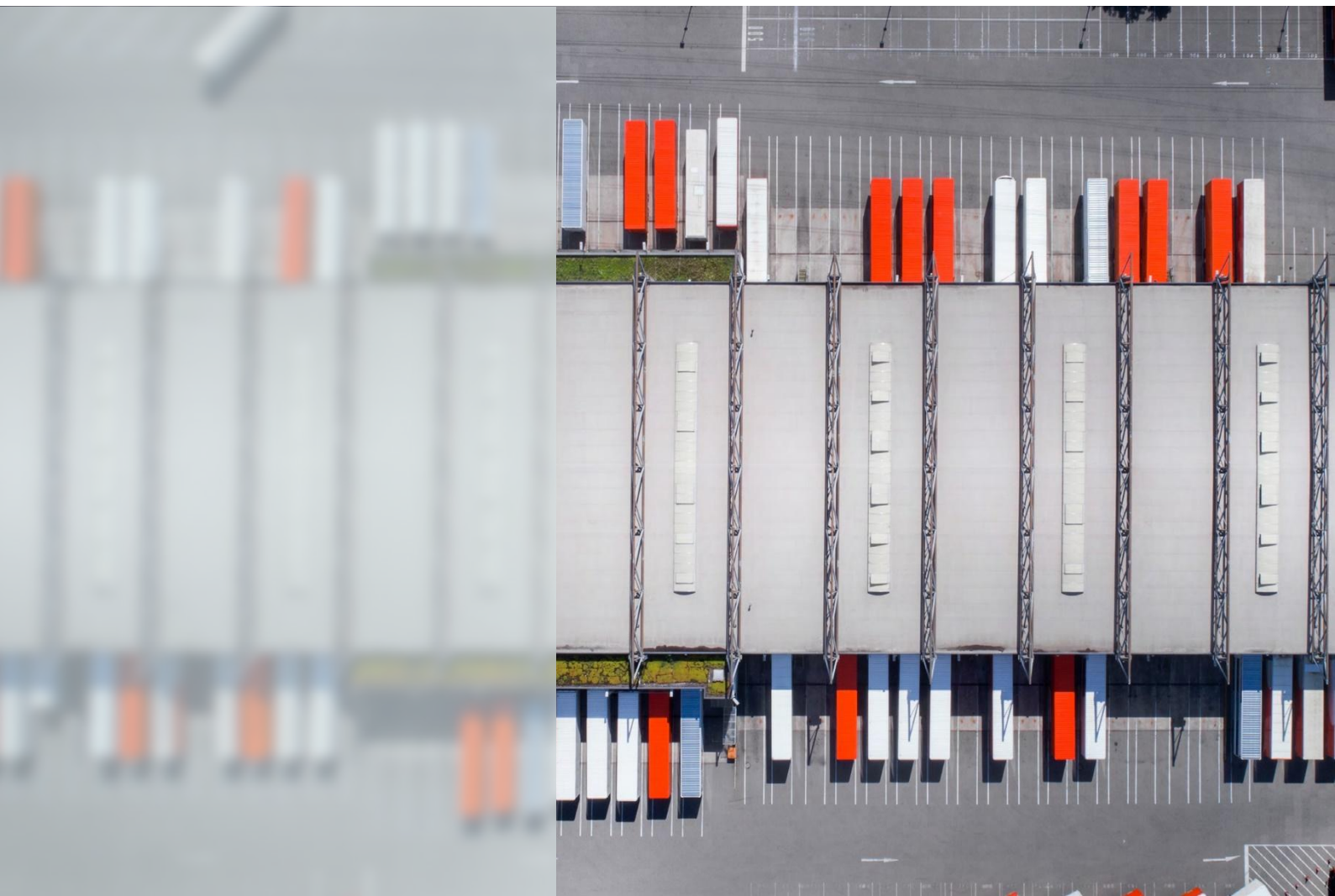


Practice Tasks

Logical Reasoning
(AP Reasoning)



How To Use This Document

General Information

This document is designed to support you in preparing for your online logical reasoning assessment.

It provides an overview of the assessment explaining what it measures and how it works.

It also provides 5 practice questions. After these practice questions, solutions and rationale have been provided to help you gain a deeper understanding. We would encourage you to attempt all the practice questions without looking at the solutions first.

Logical Reasoning

Aon's Logical Reasoning assessment measures your ability to apply given rules to logical problems in one or more steps. In the assessment, you will receive instructions along with two practice tasks that you can repeat as many times as you like before you start the actual assessment.

How Does the Logical Reasoning Assessment Work?

You are presented with a series of logical premises and a list of possible conclusions that may follow from those premises. Your task is to consider the logical implication of each premise and use them to identify the conclusion that logically follows.

To find out the right answer, you may wish to focus on finding the one answer that logically follows or systematically eliminate the alternatives that do not. Once you have identified the correct logical conclusion, you select it from the list and click the button at the bottom of the screen to proceed to the next question.

The assessment consists of 10 questions which are selected in an adaptive fashion according to your response on the previous question. The adaptive nature of this assessment means that the difficulty level may increase as you progress.

Guidance for the Practice Questions

The actual assessment you will take consists of 10 questions and you will have a limited amount of time to respond to each (2 minutes and 30 seconds per question) – you just need to work quickly and accurately, to try to get as many of the 10 questions correct as possible in the time provided.

While attempting the practice questions, you should use a pencil so you can keep track of the steps you take in working out the answers. You can then also write down your final answers, so that you can compare your answers to the solutions at the end.

To continue to practice, spend more time reviewing the example tasks and solutions provided here.

Practice Tasks

Example 1

Some of the employees who work day shifts also work double shifts.

Based on the information above, which of these statements MUST be true?

1. All employees who work day shifts also work double shifts.
2. Some employees who work day shifts do not work double shifts.
3. None of the employees who work day shifts also work double shifts.
4. Some of the employees who work double shifts do not work day shifts.
5. Some of the employees who work double shifts also work day shifts.

Example 2

There are six meeting rooms in the building: G, H, I, J, K, and L.

- I is larger than J.
- G is larger than I.
- There is one room that is smaller than G, but larger than I.
- K is larger than both G and H.

Based on the above information, which of the following is a possible ordering of the rooms from largest to smallest?

1. L, H, I, K, G, J
2. J, G, L, I, K, H
3. K, G, H, I, J, L
4. G, K, J, I, L, H
5. K, L, G, J, I, H

Example 3

A clinic has 6 nurses who begin their shift at different hours. Meal breaks must be accordingly scheduled for the 6 nurses - C, D, E, F, G, and H.

The following rules apply:

- Employee C must go on meal break before Employee F.
- Employee E must go on meal break before Employee D.
- Employee D must go on meal break before Employee C.
- Employee G must go on meal break after Employee H.
- Employee C cannot go on meal break directly before or directly after Employee H.

Given these rules, what is an acceptable order in which meal breaks can be scheduled?

1. C,E,H,G,D,F
2. H,E,G,D,C,F
3. F,E,H,D,G,C
4. E,D,H,C,F,G
5. H,E,C,D,F,G

Example 4

BrightPath Solutions has a team of consultants who conduct workshops for clients. All lead consultants have more than a year of experience. All consultants who run executive workshops have Master of Business Administration (MBA) degrees. All consultants handling multiple workshop topics are lead consultants. All consultants delivering workshops in finance have MBA degrees. Jordan runs executive workshops and handles multiple workshop topics.

Assuming these statements are true, which of the following could be FALSE?

1. Jordan has more than a year of experience.
2. Jordan has an MBA degree.
3. Jordan is a lead consultant.
4. Jordan delivers workshops in finance.
5. All consultants who handle multiple workshop topics have more than a year of experience.

Example 5

Talent coordinators at Tee-Tech are recruiting for analyst positions. While reviewing the backgrounds of current analysts, they notice that all former computer science majors became analysts. They also find that no former literature majors became analysts.

Which of the following can be concluded from the passage?

1. All former literature majors are also former computer science majors.
2. Everyone who is not a former literature major is a former computer science major.
3. Some former computer science majors are not former literature majors.
4. No one who is not a former computer science major is not a former literature major.
5. Some former literature majors are also former computer science majors.

Solutions & Rationale

Example 1 - Solution

Some of the employees who work day shifts also work double shifts.

Based on the information above, which of these statements MUST be true?

1. All employees who work day shifts also work double shifts.
2. Some employees who work day shifts do not work double shifts.
3. None of the employees who work day shifts also work double shifts.
4. Some of the employees who work double shifts do not work day shifts.
5. Some of the employees who work double shifts also work day shifts.

Answer: Option 5 - Some of the employees who work double shifts also work day shifts.

Rationale: The statement in the question tells us that "some of the employees who work day shifts also work double shifts." This means there is an overlap between employees who work day shifts and those who work double shifts. Therefore, it must be true that some of the employees who work double shifts also work day shifts, as there is a subset of employees who do both.

Example 2 - Solution

There are six meeting rooms in the building: G, H, I, J, K, and L.

- I is larger than J.
- G is larger than I.
- There is one room that is smaller than G, but larger than I.
- K is larger than both G and H.

Based on the above information, which of the following is a possible ordering of the rooms from largest to smallest?

6. L, H, I, K, G, J
7. J, G, L, I, K, H
8. K, G, H, I, J, L
9. G, K, J, I, L, H
10. K, L, G, J, I, H

Answer: Option 3 - K, G, H, I, J, L.

Rationale:

1. I is larger than J: $I > J$
2. G is larger than I: $G > I$
3. There is one room that is smaller than G, but larger than I: This room must be between G and I in size.
4. K is larger than both G and H: $K > G$ and $K > H$

From this, we can deduce the following order:

- K is the largest.
- G is larger than I.
- I is larger than J.
- There is a room between G and I in size.
- K is larger than H.

Given these conditions, let's check the options:

1. L, H, I, K, G, J - This does not fit the conditions as K should be the largest.
2. J, G, L, I, K, H - This does not fit the conditions as K should be the largest.
3. K, G, H, I, J, L - This fits all the conditions:
 - K is the largest.
 - G is larger than I.
 - I is larger than J.
 - H is smaller than K but not necessarily smaller than G.
 - L is the smallest.
4. G, K, J, I, L, H - This does not fit the conditions as K should be the largest.
5. K, L, G, J, I, H - This does not fit the conditions as L should not be larger than G.

Therefore, the correct answer is **option 3: K, G, H, I, J, L.**

Example 3 - Solution

A clinic has 6 nurses who begin their shift at different hours. Meal breaks must be accordingly scheduled for the 6 nurses - C, D, E, F, G, and H.

The following rules apply:

- Employee C must go on meal break before Employee F.
- Employee E must go on meal break before Employee D.
- Employee D must go on meal break before Employee C.
- Employee G must go on meal break after Employee H.
- Employee C cannot go on meal break directly before or directly after Employee H.

Given these rules, what is an acceptable order in which meal breaks can be scheduled?

1. C,E,H,G,D,F
2. H,E,G,D,C,F
3. F,E,H,D,G,C
4. E,D,H,C,F,G
5. H,E,C,D,F,G

Answer: Option 2 - H,E,G,D,C,F

Rationale:

Rules Recap:

1. C before F
2. E before D
3. D before C
4. G after H
5. C cannot be directly before or after H

Let's Analyze Option 2: H, E, G, D, C, F

Position:

1. H
2. E
3. G
4. D
5. C
6. F

Now test the rules:

1. C before F?
 - C is 5th, F is 6th → Yes.
2. E before D?
 - E is 2nd, D is 4th → Yes.
3. D before C?
 - D is 4th, C is 5th → Yes.
4. G after H?
 - G is 3rd, H is 1st → Yes.
5. C not directly before/after H?
 - H is 1st, C is 5th → Not adjacent → Yes.

Only **Option 2** satisfies all 5 rules. Therefore, it is the correct answer.

Example 4 - Solution

BrightPath Solutions has a team of consultants who conduct workshops for clients. All lead consultants have more than a year of experience. All consultants who run executive workshops have Master of Business Administration (MBA) degrees. All consultants handling multiple workshop topics are lead consultants. All consultants delivering workshops in finance have MBA degrees. Jordan runs executive workshops and handles multiple workshop topics.

Assuming these statements are true, which of the following could be FALSE?

1. Jordan has more than a year of experience.
2. Jordan has an MBA degree.
3. Jordan is a lead consultant.
4. Jordan delivers workshops in finance.
5. All consultants who handle multiple workshop topics have more than a year of experience.

Answer: Option 4 - Jordan delivers workshops in finance.

Rationale:

What can we conclude about Jordan based on the statements provided?

- Jordan handles multiple workshop topics, so per statement 3, Jordan is a lead consultant.
- As a lead consultant, Jordan must have more than a year of experience (from statement 1).
- Jordan runs executive workshops, so Jordan must have an MBA degree (statement 2).

So, Jordan:

- Has more than a year of experience
- Has an MBA degree
- Is a lead consultant

However, **nothing in the information provided confirms that Jordan works specifically in finance.**

- Based on the information, Jordan runs executive workshops and works on multiple topics.
- It is also true that anyone who delivers workshops in finance must have an MBA, and Jordan has one—but having an MBA doesn't mean you automatically work in finance.
- So it's possible that Jordan doesn't teach anything in finance at all.

Therefore, this is the only statement that's **not guaranteed to be true** based on the facts provided.

Example 5 - Solution

Talent coordinators at Tee-Tech are recruiting for analyst positions. While reviewing the backgrounds of current analysts, they notice that all former computer science majors became analysts. They also find that no former literature majors became analysts.

Which of the following can be concluded from the passage?

1. All former literature majors are also former computer science majors.
2. Everyone who is not a former literature major is a former computer science major.
3. Some former computer science majors are not former literature majors.
4. No one who is not a former computer science major is not a former literature major.
5. Some former literature majors are also former computer science majors.

Answer: Option 3 - Some former computer science majors are not former literature majors.

Rationale:

What We Know from the Passage:

1. All former computer science majors became analysts.
→ So, if someone studied computer science, they definitely became an analyst.
2. No former literature majors became analysts.
→ So, if someone studied literature, they definitely did not become an analyst.

What Can Be Concluded?

Option 3 - "Some former computer science majors are not former literature majors."

This can be true based on the information given. Here's why:

- The statement says "some," which only means "at least one."
- Since all computer science majors became analysts, and no literature majors became analysts, there must be at least some computer science majors who didn't also study literature—otherwise, how would they qualify as analysts while literature majors don't?
- There's no overlap allowed between analysts and literature majors. So at least some (likely most) computer science majors had nothing to do with literature.

Thus, **option 3** is a logically safe conclusion based on the facts provided.